

Page 1: Preliminary Analysis & Project Initiation

The first stage of any system development is **preliminary analysis**. This phase aims to identify customer needs, evaluate the system concept for feasibility, perform economic and technical analysis, and create a system definition that serves as the foundation for all subsequent engineering work.

1.1. Request Clarification

Initially, when a customer requests a system, they are often unsure of the exact requirements. To bridge this gap:

- The analyst personally meets with the end-user or customer to extract maximum information.
- The analyst uses tact and communication skills to define exact

requirements.

- All gathered information is recorded in the **System Concept Document**.

1.2. The Feasibility Study

After clarifying the request, the analyst proposes solutions and checks their practicality through a feasibility study.

- **Technical Feasibility:** Considers if the required technology and resources (manpower like programmers/testers, software, and hardware) are available.
- **Economic Feasibility:** Determines if the expected benefits equal or exceed costs. It evaluates the effectiveness of the proposed system through cost-benefit analysis.
- **Legal Feasibility:** Includes studying contracts, liability, and potential legal violations that might be unknown to technical staff.
- **Operational Feasibility:** Determines if

the system will actually be used and if there will be user resistance.

Page 2: Cost-Benefit Analysis (CBA) & Categories

Developing an IT application is an investment that carries risks, as estimates can be wrong. CBA helps management visualize costs, benefits, and risks.

2.1. Classifying Costs and Benefits

To perform an accurate CBA, costs and benefits are categorized as follows:

- **Tangible vs. Intangible:** Tangible items like hardware and salaries can be measured. Intangible items, such as customer satisfaction or the cost of a system breakdown during banking hours, cannot be easily measured but must be considered.

- **Direct vs. Indirect:** Direct costs have a specific value associated with the project, while indirect costs result from operations not directly tied to the system, such as insurance, maintenance, or air conditioning.
- **Fixed vs. Variable:** Fixed costs like hardware depreciation do not change. Variable costs, such as paper or disks, vary based on the amount of use.

2.2. Evaluation Methods

Several methods exist to evaluate financial data:

- **Present Value Analysis:** Calculates costs and benefits in terms of today's value for long-term projects.
- **Net Present Value (NPV):** Calculated as Benefits minus Costs; it is often expressed as a percentage of the investment.
- **Break-even Analysis:** Finds the point

where the cost of the proposed system equals the cost of the current system.

- **Cash-flow Analysis:** Keeps track of accumulated costs and revenues on a regular basis.

Page 3: Software Estimation Techniques

Estimation is the process of predicting the software size and cost. Because software is the most expensive element in many systems, large errors can lead to disastrous financial losses.

3.1. Decomposition Techniques

These involve breaking the project into major functions to estimate cost and effort stepwise:

